

A ADDITIONAL PROOFS OF LEMMAS AND THEOREMS

THEOREM 3.1. Given a graph $G = (V, E)$, and two vertex sets $R, A \subseteq V$. In FDP, when

$$T > O\left(\frac{\Delta(G)(\text{vol}(R) + |A|\Delta(G)^2)}{\epsilon^2}\right),$$

the following convergence guarantee holds: $\|\mathbf{r} + \boldsymbol{\beta}\|_\infty - \rho_R^* \leq \epsilon$, where $\text{vol}(R)$ is the sum of degrees in G for all vertices in the set R , and $\Delta(G)$ denotes the maximum degree of graph G .

PROOF. In the t -th iteration, for simplify, we let $\mathbf{z}^{(t)} = \mathbf{r}^{(t)} + \boldsymbol{\beta}^{(t)}$, and $\mathbf{s}^{(t)} = \mathbf{z}^{(t)} - \mathbf{z}^{(t-1)}$, we have:

$$\frac{\mathbf{z}^{(t)}}{t} = \left(1 - \frac{1}{t}\right) \cdot \frac{\mathbf{z}^{(t-1)}}{t-1} + \frac{1}{t} \cdot \mathbf{s}^{(t-1)}$$

Clearly, this update can be interpreted as a standard Frank-Wolfe step with a learning rate of $\gamma_t = \frac{1}{t}$. Now, consider the function $f(\mathbf{z}) = (\|\mathbf{z}\|)^2$, clearly, the $\min_{\mathbf{z} \in C} f(\mathbf{z})$ is exactly the optimal solution of QP(G, A), where C denotes the set of all feasible vectors $\mathbf{z}$ for QP(G, A).

Then, we can introduce the following theorem to analyze the convergence rate of Algorithm 1.

THEOREM A.1 ([37]). Let $\mathbf{z}^*$ denote the optimal solution of $\min_{\mathbf{z} \in C} f(\mathbf{z})$. For each $t \geq 1$ in Algorithm 1, the iterates of $\mathbf{x}^{(t)}$ satisfies:

$$f(\mathbf{z}^{(t)}) - f(\mathbf{z}^*) \leq \gamma_t \cdot \xi \cdot (1 + \delta),$$

where ξ is the curvature constant of $f(\mathbf{z})$, and δ is the error in the linear minimization step.

Now, we analyze the curvature constant ξ and the accuracy δ of the objective function:

$$\min f(\mathbf{z}) = \sum_{u \in V} (r(u) + \beta(u))^2 = \|\mathbf{z}\|^2$$

We can easily derive $\delta = 0$, and the upper bound of ξ can be calculated by the following theorem.

THEOREM A.2. The curvature constant ξ of $f(\mathbf{z}) = \sum_{u \in V} (r(u) + \beta(u))^2$ satisfies $\xi \leq 8(\Delta(G) + 1) \cdot (|A|\Delta(G)^2 + \text{vol}(R))$.

PROOF. Denote C by the set of $\mathbf{x}$ of all feasible solutions for QP(G, A). By the definition of the curvature constant, we have

$$\xi = \sup_{\mathbf{x}_1, \mathbf{y} \in C, \gamma \in [0, 1], \mathbf{x}_2 = \mathbf{x}_1 + \gamma(\mathbf{y} - \mathbf{x}_1)} \frac{2}{\gamma^2} (f(\mathbf{x}_2) - f(\mathbf{x}_1) - \nabla f(\mathbf{x}_1)^T (\mathbf{x}_2 - \mathbf{x}_1))$$

The gradient of $f(\mathbf{x})$ is $\nabla f(\mathbf{x})_{u,v} = 2x_u$. Therefore,

$$\begin{aligned} & f(\mathbf{x}_2) - f(\mathbf{x}_1) - \nabla f(\mathbf{x}_1)^T (\mathbf{x}_2 - \mathbf{x}_1) \\ &= \sum_{u \in V} (x_2(u)^2 - x_1(u)^2) - \sum_{u \in V} 2x_1(u)(x_2(u) - x_1(u)) \\ &= \sum_{u \in V} (x_1(u) - x_2(u))^2 \\ &= \|\mathbf{x}_1 - \mathbf{x}_2\|^2 \\ &= \gamma^2 \|\mathbf{y} - \mathbf{x}_1\|^2 \\ &\leq \gamma^2 C^2 \end{aligned}$$

where $\mathfrak{R}$ is the diameter of C :

$$\begin{aligned} \mathfrak{R}^2 &= \sup_{\mathbf{x} \in C, \mathbf{x} = \mathbf{r} + \boldsymbol{\beta}} \sum_{u \in V} (r(u) + \beta(u))^2 \\ &\leq \sup_{\mathbf{x} \in C, \mathbf{x} = \mathbf{r} + \boldsymbol{\beta}} \sum_{u \in V} \left(\sum_{v \in N(u)} \alpha_{u,v} + \beta(u) \right)^2 \\ &= \sup_{\mathbf{x} \in C, \mathbf{x} = \mathbf{r} + \boldsymbol{\beta}} \sum_{u \in V} (d(u) + 1) \left(\beta(u)^2 + \sum_{v \in N(u)} (\alpha_{u,v})^2 \right) \\ &\leq (\Delta(G) + 1) \sup_{\mathbf{x} \in C, \mathbf{x} = \mathbf{r} + \boldsymbol{\beta}} \sum_{u \in V} \left(\beta(u)^2 + \sum_{v \in N(u)} (\alpha_{u,v})^2 \right) \\ &= (\Delta(G) + 1) \sup_{\mathbf{x} \in C, \mathbf{x} = \mathbf{r} + \boldsymbol{\beta}} \left(\sum_{u \in A} \beta(u)^2 + \sum_{u \in V} \sum_{v \in N(u)} \alpha_{u,v}^2 \right) \\ &\leq (\Delta(G) + 1)(4|A|\Delta(G)^2 + 4\text{vol}(R)) \end{aligned}$$

That is, $\xi \leq 8(|A|\Delta(G)^3 + \text{vol}(R)\Delta(G))$ $\square$

Now, at the T -th iteration, for the objective function $f(\mathbf{z})$, we have achieved the following bound:

$$\begin{aligned} f(\mathbf{z}^{(T)}) - f(\mathbf{z}^*) &\leq \gamma_T \cdot \xi \\ &\leq \frac{1}{T} 8(|A|\Delta(G)^3 + \text{vol}(R)\Delta(G)) \end{aligned}$$

To ensure $f(\mathbf{z}^{(T)}) - f(\mathbf{z}^*) \leq \epsilon$, we need to show:

$$\frac{8(|A|\Delta(G)^3 + \text{vol}(R)\Delta(G))}{T} \leq \epsilon^2$$

In other words,

$$T \geq \frac{8(|A|\Delta(G)^3 + \text{vol}(R)\Delta(G))}{\epsilon^2}$$

Absorbing constant factors into the asymptotic notation, this yields the final complexity bound:

$$T = O\left(\frac{|A|\Delta(G)^3 + \text{vol}(R)\Delta(G)}{\epsilon^2}\right)$$

This completes the proof. $\square$

THEOREM 4.1. Given a graph $G = (V, E)$ and two vertex sets $R, A \subseteq V$, we have $\text{OPT}(\text{QP}(G, A)) = \text{OPT}(\text{QP}'(G, A))$.

PROOF. Let

$$g(\mathbf{r}, \boldsymbol{\beta}) = \sum_{u \in V} (r(u) + \beta(u))^2, \quad z(\mathbf{r}) = |A| \left(\max_{v \in V} r(v) \right)^2 + \sum_{u \in V \setminus A} r(u)^2,$$

denote the objective functions of QP(G, A) and QP'(G, A), respectively. We prove the theorem in two steps.

(i) $\text{OPT}(\text{QP}(G, A)) \geq \text{OPT}(\text{QP}'(G, A))$. The program $\text{QP}'(G, A)$ can be considered as a relaxation of $\text{QP}(G, A)$ that removes only the constraints related to β . Therefore, any feasible triple $(\mathbf{r}, \beta, \alpha)$ of $\text{QP}(G, A)$ implies that the pair $(\mathbf{r}, \alpha)$ is a feasible solution to $\text{QP}'(G, A)$. Moreover,

$$\begin{aligned} g(\mathbf{r}, \beta) &= \sum_{u \in V} (r(u) + \beta(u))^2 \\ &\geq |A| \cdot \left(\max_{v \in V} (r(v) + \beta(v)) \right)^2 + \sum_{u \in V \setminus A} (r(u) + \beta(u))^2 \\ &\geq z(\mathbf{r}). \end{aligned}$$

(ii) $\text{OPT}(\text{QP}(G, A)) \leq \text{OPT}(\text{QP}'(G, A))$. Let $(\mathbf{r}^*, \alpha^*)$ be an optimal pair for $\text{QP}'(G, A)$ and set $\tau = \max_{v \in V} r^*(v)$. Define

$$\beta^*(u) = \begin{cases} \tau - r^*(u), & u \in A, \\ 0, & u \in V \setminus A. \end{cases}$$

Then $r^*(u) + \beta^*(u) = \tau$ for all $u \in A$ and $r^*(v) + \beta^*(v) \leq \tau$ for $v \notin A$; hence $(\mathbf{r}^*, \beta^*, \alpha^*)$ is a feasible solution for $\text{QP}(G, A)$. Its objective value is

$$\begin{aligned} g(\mathbf{r}^*, \beta^*) &= \sum_{u \in V} (r^*(u) + \beta^*(u))^2 \\ &= |A| \tau^2 + \sum_{u \in V \setminus A} r^*(u)^2 = z(\mathbf{r}^*) \leq \text{OPT}(\text{QP}'(G, A)). \end{aligned}$$

Combining (i) and (ii), we obtain $\text{OPT}(\text{QP}(G, A)) = \text{OPT}(\text{QP}'(G, A))$, as claimed. $\square$

THEOREM 4.3. The Lipschitz constant L_f of $\nabla f(\alpha)$, is bounded by $2 \cdot (|A| + 1) \cdot \sqrt{\Delta(G) \cdot \text{vol}(R)}$.

$$L_f^2 = \|\nabla f(\alpha)\|^2 = \sum_{(u,v) \in E} \sum_{(x,y) \in E} \left(\frac{\partial (\nabla f(\alpha)_{u,v})}{\partial \alpha_{x,y}} \right)^2$$

We have:

$$\frac{\partial (\nabla f(\alpha)_{u,v})}{\partial \alpha_{x,y}} = \frac{\partial (2 \cdot (\mathbb{1}[u \notin A] + g_{u,v}) \cdot r(u))}{\partial \alpha_{x,y}} \leq \begin{cases} 2 \cdot (|A| + 1), & u = x \\ 0, & u \neq x \end{cases}$$

Therefore,

$$\begin{aligned} L_f^2 &\leq \sum_{u \in V} \sum_{(u,v) \in E} \sum_{(u,y) \in E} (2 \cdot (|A| + 1))^2 \\ &\leq \sum_{u \in V} \sum_{(u,v) \in E} 4(|A| + 1)^2 \cdot d(u) \\ &\leq 4(|A| + 1)^2 \Delta(G) \sum_{u \in V} \sum_{(u,v) \in E} 1 \\ &\leq 4(|A| + 1)^2 \Delta(G) \cdot \text{vol}(R) \end{aligned}$$

Therefore, $L_f \leq 2(|A| + 1) \sqrt{\Delta(G) \cdot \text{vol}(R)}$.

THEOREM 4.4. Given a graph $G = (V, E)$, and two vertex sets $R, A \subseteq V$. In PGD-ADS, when

$$T > O \left(\frac{\sqrt{|A|} \cdot \text{vol}(R)^{\frac{1}{4}} \cdot \Delta(G)^{\frac{3}{4}} \cdot \sqrt{\text{vol}(R) + |A| \Delta(G)}}{\epsilon} \right),$$

the following convergence guarantee holds: $\|\lambda\|_\infty - \rho_R^* \leq \epsilon$.

PROOF. We first provide our proof roadmap: (1) we prove that if $f(\alpha) - f(\alpha^*) \leq \mu$, and we have $\|\lambda\|_\infty - \|\lambda^*\|_\infty \leq \sqrt{\mu}$, and (2) Let $\mu := \frac{2 \cdot \|x^{(0)} - x^*\|^2}{\eta \cdot T^2} = \epsilon^2$. Then, by Theorem 4.2, we establish the relationship between the target accuracy ϵ and the number of iterations T , thus completing the proof.

Firstly, $f(\alpha) - f(\alpha^*)$ can be rewritten as:

$$\|\lambda\|^2 - \|\lambda^*\|^2.$$

For each $u \in V$, let $\delta(u) = \lambda(u) - \lambda^*(u)$, we have:

$$\begin{aligned} \|\lambda\|^2 - \|\lambda^*\|^2 &= \sum_{u \in V} \lambda(u)^2 - \lambda^*(u)^2 \\ &= \sum_{u \in V} (2\lambda^*(u) \cdot \delta(u) + \delta(u)^2) \leq \mu. \end{aligned}$$

Afterwards, we aim to prove that:

$$\sum_{u \in V} (2\lambda^*(u) \cdot \delta(u)) \geq 0,$$

thus $\sum_{u \in V} \delta(u)^2 \leq \mu$, in other words, $(\|\lambda - \lambda^*\|)^2 \leq \mu$.

To achieve this goal, we need to establish two results: (1) show that $\lambda^*(u)$ is identical for all $u \in V$; (2) prove that $\sum_{u \in V} \delta(u) \geq 0$.

(1) For the first item, by Fujishige's theorem [29], when the feasible region is the base contrapolymatroid of a supermodular function, the optimal solution to the ℓ_2 -norm minimization assigns identical values to all coordinates. A set function $g : 2^{\mathcal{U}} \rightarrow \mathbb{R}$ is said to be *supermodular* if for any subsets $X, Y \subseteq \mathcal{U}$, it satisfies: $g(X) + g(Y) \leq g(X \cup Y) + g(X \cap Y)$.

Let $g : 2^{\mathcal{U}} \rightarrow \mathbb{R}^+$ be a supermodular function, and $\vec{x} \in \mathbb{R}^{\mathcal{U}}$ be a nonnegative vector. The *base contrapolymatroid* defined by g is:

$$B_g = \{ \vec{x} \in \mathbb{R}^{\mathcal{U}} \mid \vec{x} \geq 0, \vec{x}(S) \geq g(S), \forall S \subseteq \mathcal{U}, x(\mathcal{U}) = g(\mathcal{U}) \},$$

where $\forall S \subseteq \mathcal{U}, \vec{x}(S) = \sum_{u \in S} x(u)$. Intuitively, the base contrapolymatroid B_g can be viewed as the feasible region for a $|\mathcal{U}|$ -dimensional vector under a set of lower bounds and a single linear equality constraint.

Clearly, $\forall S \subseteq V$, let $f(S) = 2|E(S)| - \sum_{u \in S \setminus R} |N(u, G[S])|$, f is a supermodular function, so that the constraints of $\text{QP}'(G, A)$ can be compactly written as:

$$\lambda(S) \geq f(S), \quad \forall S \subseteq V, \quad \lambda(V) = f(V),$$

where $\lambda(S) = \sum_{u \in S} \lambda(u)$.

The *base contrapolymatroid* of $f(S)$ is:

$$B_f = \{ \lambda \in \mathbb{R}^n \mid \lambda \geq 0, \lambda(S) \geq f(S), \forall S \subseteq V \},$$

where $\lambda(V) = f(V)$.

Based on the B_f , and vector λ we can rewrite the $\text{QP}'(G, A)$:

$$\text{QP}'(G, A) : \quad \min_{\lambda \in B_f} \sum_{u \in V} \lambda(u)^2.$$

Thus, by Fujishige's theorem [29], the optimal solution of $\text{QP}'(G, A)$ is the lexicographically minimal base $\lambda^* \in B_f$. That is, it follows that $\lambda^*(u)$ is identical for all $u \in V$.

(2) As $\lambda \in B_f$ and $\lambda(V) = f(V)$, we have $\sum_{u \in V} \delta(u) = \lambda(V) - \lambda^*(V) \geq 0$.

By combining (1) and (2), we include $(\|\lambda - \lambda^*\|)^2 \leq \mu$. Let $\mu = \epsilon^2$, we have $\|\lambda - \lambda^*\| \leq \epsilon$, on the other hands, we also have:

$$\|\lambda\|_\infty - \|\lambda^*\|_\infty \leq \|\lambda - \lambda^*\|_\infty \leq \|\lambda - \lambda^*\| \leq \epsilon.$$

In other words, if we have $f(\alpha) - f(\alpha^*) \leq \epsilon^2$, we can finish the proof. Based on the Theorem 4.2, we have:

$$f(\alpha) - f(\alpha^*) \leq \frac{2 \cdot \|\lambda^{(0)} - \lambda^*\|^2}{\eta \cdot T^2}$$

The gap of $\|\lambda^{(0)} - \lambda^*\|^2$ is graduated by the following:
As $\|\lambda^{(0)} - \lambda^*\| \leq \|\lambda^{(0)}\|$, we consider:

$$\begin{aligned} \|\lambda^{(0)}\|^2 &= \sum_{u \in V} \lambda(u)^2 = |A| \cdot (\max_{v \in V} r(v))^2 + \sum_{u \in V} \left(\sum_{(u,v) \in E \wedge (u,v) \notin A} \alpha_{u,v} \right)^2 \\ &\leq |A| \cdot 4 \cdot \Delta(G)^2 + \sum_{u \in V} d(u, G) \left(\sum_{(u,v) \in E} \alpha_{u,v}^2 \right) \\ &= |A| \cdot 4 \cdot \Delta(G)^2 + \Delta(G) \cdot \left(\sum_{(u,v) \in E \wedge (u < v)} \alpha_{u,v}^2 + \alpha_{v,u}^2 \right) \\ &\leq |A| \cdot 4 \cdot \Delta(G)^2 + \Delta(G) \cdot \left(\sum_{(u,v) \in E} w_{u,v}^2 \right) \\ &\leq 4 \cdot |A| \cdot \Delta(G)^2 + 4\Delta(G) \cdot \text{vol}(R) \end{aligned}$$

Therefore, $\|\lambda^{(0)} - \lambda^*\| \leq 2\sqrt{|A| \cdot \Delta(G)^2 + \text{vol}(R) \cdot \Delta(G)}$.
To ensure $f(\alpha) - f(\alpha^*) \leq \epsilon^2$, we need to show:

$$\frac{2 \cdot (4 \cdot |A| \cdot \Delta(G)^2 + 4 \cdot \text{vol}(R) \cdot \Delta(G))}{\frac{1}{2 \cdot (|A|+1) \cdot \sqrt{\Delta(G) \cdot \text{vol}(R)}} \cdot T^2} \leq \epsilon^2$$

In other words,

$$T^2 \geq \frac{16 \cdot (\text{vol}(R) \cdot \Delta(G) + |A| \cdot \Delta(G)^2) \cdot (|A| + 1) \cdot \sqrt{\Delta(G) \cdot \text{vol}(R)}}{\epsilon^2}$$

Absorbing constant factors into the asymptotic notation, this yields the final complexity bound:

$$T = O\left(\frac{\sqrt{|A|} \cdot \text{vol}(R)^{\frac{1}{4}} \cdot \Delta(G)^{\frac{3}{4}} \cdot \sqrt{\text{vol}(R) + |A|\Delta(G)}}{\epsilon}\right)$$

This completes the proof. $\square$

THEOREM 5.3 Given a graph $G = (V, E)$, two vertex sets R and A , and the $[\rho_R^*]$ -r-core, $\mathcal{M}^*$, the ADS $\mathcal{D} = G[S^*]$ must be contained within $G[V(\mathcal{M}^*) \cup A]$, where ρ_R^* is the density of $\mathcal{D}$.

PROOF. We prove this by contradiction. Suppose that the ADS $\mathcal{D} = G[S^*]$ is not fully contained in $G[V(\mathcal{H}^*) \cup A]$. Then there exists a vertex $u \in S^* \wedge u \notin A$ such that:

$$d_R(u, G[S^*]) < \rho_R^*.$$

Consider removing u from S^* , and let $S' = S^* \setminus \{u\}$. Since $\mathcal{D}$ is an ADS, it holds that $\rho_R(G[S']) \leq \rho_R^*$. On the other hand, we can compute $\rho_R(G[S'])$ as:

$$\rho_R(G[S']) = \frac{2|E(S')|}{|S'|} = \frac{2|E(S^*)| - 2d_R(u, G[S^*])}{|S^*| - 1},$$

and since $d_R(u, G[S^*])$ counts the number of R -neighbors of u in $G[S^*]$, we further have:

$$2|E(S')| = \rho_R^* \cdot |S^*| - d_R(u, G[S^*]).$$

Therefore,

$$\rho_R(G[S']) = \frac{\rho_R^* \cdot |S^*| - d_R(u, G[S^*])}{|S^*| - 1}.$$

Since $\rho_R(G[S']) \leq \rho_R^*$, we obtain:

$$\begin{aligned} \frac{\rho_R^* \cdot |S^*| - d_R(u, G[S^*])}{|S^*| - 1} &\leq \rho_R^* \\ \rho_R^* \cdot |S^*| - d_R(u, G[S^*]) &\leq \rho_R^* \cdot (|S^*| - 1) \\ d_R(u, G[S^*]) &\geq \rho_R^*. \end{aligned}$$

This contradicts the assumption that $d_R(u, G[S^*]) < \rho_R^*$. Hence, we have $\forall u \in S^* \setminus A$, $d_R(u, G[S^*]) \geq \rho_R^*$. By the definition of the k -r-core, the theorem holds. $\square$

LEMMA 5.5. Given a graph $G = (V, E)$, and its subgraph S , we have: $\sum_{v \in V(S)} d_R(v, S) = 2 \times (2|E(S)| - \sum_{u \in V(S) \setminus R} |N(u, S)|)$.

PROOF. To prove this lemma, we first have:

$$\begin{aligned} 2|E(S)| - \sum_{u \in V(S) \setminus R} |N(u, S)| &= \sum_{u \in V(S)} d(u, S) - \sum_{u \in V(S) \wedge u \notin R} |N(u, S)| \\ &= \sum_{u \in V(S) \wedge u \in R} d(u, S) + \sum_{u \in V(S) \wedge u \notin R} d(u, S) - \sum_{u \in V(S) \wedge u \notin R} |N(u, S)| \\ &= \sum_{u \in V(S) \wedge u \in R} d(u, S). \end{aligned}$$

On the other hand, we have:

$$\begin{aligned} \sum_{u \in V(S)} d_R(u, S) &= \sum_{u \in V(S) \wedge u \in R} d_R(u, S) + \sum_{u \in V(S) \wedge u \notin R} d_R(u, S) \\ &= \sum_{u \in V(S) \wedge u \in R} d(u, S) + d(u, G[V(S) \cap R]) + \sum_{u \in V(S) \wedge u \notin R} d(u, G[V(S) \cap R]) \\ &= \sum_{u \in V(S) \wedge u \in R} d(u, S) + \sum_{u \in V(S)} d(u, G[V(S) \cap R]) \\ &= \sum_{u \in V(S) \wedge u \in R} d(u, S) + \sum_{(u,v) \in E(S)} \mathbb{1}[u \in R] \\ &= \sum_{u \in V(S) \wedge u \in R} d(u, S) + \sum_{v \in V(S) \wedge v \in R} d(v, S) \end{aligned}$$

Thus, the above lemma holds. $\square$

LEMMA 5.6. Given a graph $G = (V, E)$ and two vertex sets $R, A \subseteq V$, if there exists a k -r-core $\mathcal{M}$ in G , then the NR-density of the ADS in G is at least $\frac{k^2}{2(k+|A|)}$.

PROOF. Consider the subgraph $G[V(\mathcal{M}) \cup A]$, it's NR-density satisfies:

$$\begin{aligned} \rho_R(G[V(\mathcal{M}) \cup A]) &= \frac{\sum_{u \in V(\mathcal{M}) \cup A} d_R(u, G[V(\mathcal{M}) \cup A])}{2 \cdot |V(\mathcal{M}) \cup A|} \\ &\geq \frac{\sum_{u \in V(\mathcal{M})} d_R(u, \mathcal{M})}{2 \cdot (|V(\mathcal{M})| + |A|)} \\ &\geq \frac{k \cdot |V(\mathcal{M})|}{2 \cdot (|V(\mathcal{M})| + |A|)}. \end{aligned}$$

Consider the function $f(x) = \frac{k \cdot x}{x + |A|}$. Its derivative is given by

$$f'(x) = \frac{k \cdot |A|}{(x + |A|)^2} > 0,$$

which indicates that $f(x)$ is strictly increasing with respect to x .

Now, let $x = |V(\mathcal{M})|$. Since $|V(\mathcal{M})| \geq k$, we have:

$$\rho_R(G[V(\mathcal{M}) \cup A]) \geq \frac{k^2}{2 \cdot (k + |A|)}.$$

Therefore, $\frac{k^2}{2 \cdot (k + |A|)}$ is a lower bound on the NR-density of G . $\square$

THEOREM 6.1. Given a graph $G = (V, E)$, and two vertex sets $R, A \subseteq V$. In PASTA, when

$$T > O\left(\frac{\sqrt{|A|} \cdot \Delta(G) \cdot \sqrt{(\text{vol}(R) + |A|\Delta(G))}}{\epsilon}\right),$$

the algorithm is guaranteed to return a $(1 + \epsilon)$ -approximation solution.

The result in Theorem 6.1 is straightforward, since PASTA is built upon PGD-ADS, and thus inherits its convergence properties. Specifically, the convergence rate of our method is significantly better than that of FDP, as shown in the following theorem:

THEOREM 6.2. To find a $(1 + \epsilon)$ -approximation solution, PASTA reduces the overall time complexity at least by $O\left(\frac{\text{vol}(R)^{\frac{3}{4}} \cdot \Delta(G)^{\frac{5}{4}}}{\epsilon}\right)$ over FDP, where all variables keep the same meanings as Theorem 4.4

PROOF. To obtain a $(1 + \epsilon)$ -approximation solution, the baseline method FDP and PASTA take $O\left(\frac{(\Delta(G) \cdot (\text{vol}(R) + |A|\Delta(G)^2)) \cdot \text{vol}(R)}{\epsilon^2}\right)$, and $O\left(\frac{\sqrt{|A|} \cdot \text{vol}(R)^{\frac{1}{4}} \cdot \Delta(G)^{\frac{3}{4}} \cdot \sqrt{(\text{vol}(R) + |A|\Delta(G))}}{\epsilon}\right)$ time theoretically, respectively. Compared to FDP, our method achieves a reduction of:

$$O\left(\frac{\text{vol}(R)^{\frac{3}{4}} \Delta(G)^{\frac{1}{4}} (\text{vol}(R) + |A|\Delta(G)^2)}{\epsilon \cdot \sqrt{|A|} \sqrt{\text{vol}(R) + |A|\Delta(G)}}\right),$$

which can be further reduced to $O\left(\frac{\text{vol}(R)\Delta(G)}{\epsilon}\right)$. This implies that the function $g(x)$ (where $x = |A|$) should be defined as:

$$g(x) = \frac{\text{vol}(R)^{\frac{3}{4}} \Delta(G)^{\frac{1}{4}} (\text{vol}(R) + x\Delta(G)^2)}{\sqrt{x} \sqrt{\text{vol}(R) + x\Delta(G)}}$$

To find the minimum of $g(x)$, we can equivalently minimize $g(x)^2$ as the square root is a monotonic function. We can rewrite $g(x)^2$ (ignoring the constant $\Delta(G)$ for minimization purposes):

$$\frac{g(x)^2}{\text{vol}(R)^{\frac{3}{2}} \Delta(G)^{\frac{1}{2}}} = \frac{(\text{vol}(R) + x\Delta(G)^2)^2}{x(\text{vol}(R) + x\Delta(G))}$$

Let $h(x) = \frac{(\text{vol}(R) + x\Delta(G)^2)^2}{x(\text{vol}(R) + x\Delta(G))}$. Let $a = \text{vol}(R)$, $b = \Delta(G)$, we can rewrite $h(x)$ as $h(x) = \frac{(a + b^2x)^2}{x(a + bx)}$. Now, we compute the derivative of $h(x)$ with respect to x :

$$h'(x) = \frac{d}{dx} \left(\frac{(a + b^2x)^2}{x(a + bx)} \right) = \frac{a(a + b^2x)}{x^2(a + bx)^2} ((b^2 - 2b)x - a)$$

Given that $a, b > 0$ and $x \in [0, |R|]$, there are two cases regarding whether $0 < b \leq 2$ or $b > 2$.

When $0 < b \leq 2$, $h'(x) < 0$ when $x \in [0, |R|]$. Therefore, the minimum value of $h(x)$ (thus $g(x)$) occurs at $x = |R|$. Note that $|R| < \text{vol}(R) = a$. That is, we have $g(x) \geq g(|R|) \geq g(a) = \frac{a^{\frac{3}{4}} b^{\frac{1}{4}} (a + b^2 \cdot a)}{\sqrt{a(a + b \cdot a)}} =$

$$\frac{a^{\frac{3}{4}} b^{\frac{1}{4}} (1 + b^2)}{\sqrt{1 + b}} = \frac{\text{vol}(R)^{\frac{3}{4}} \Delta(G)^{\frac{1}{4}} (1 + \Delta(G)^2)}{\sqrt{1 + \Delta(G)}} = O(\text{vol}(R)^{\frac{3}{4}} \Delta(G)^{\frac{7}{4}}).$$

When $b > 2$, we have

$$h'(x) \begin{cases} < 0, & x \in [0, \frac{a}{b(b-2)}] \\ > 0, & x \in (\frac{a}{b(b-2)}, |R|] \end{cases}$$

Therefore, the minimum value of $h(x)$ (thus $g(x)$) occurs at $x = \frac{a}{b(b-2)}$. That is, we have

$$\begin{aligned} g(x) &\geq g\left(\frac{a}{b(b-2)}\right) \\ &= \frac{a^{\frac{3}{4}} b^{\frac{1}{4}} \left(a + b^2 \frac{a}{b(b-2)}\right)}{\sqrt{\frac{a}{b(b-2)} \left(a + b \cdot \frac{a}{b(b-2)}\right)}} \\ &= 2a^{\frac{3}{4}} b^{\frac{1}{4}} \sqrt{b(b-1)} \\ &= 2 \left(\text{vol}(R)^{\frac{3}{4}} \Delta(G)^{\frac{1}{4}} \sqrt{\Delta(G)(\Delta(G) - 1)}\right) \\ &= O\left(\text{vol}(R)^{\frac{3}{4}} \Delta(G)^{\frac{5}{4}}\right) \end{aligned}$$

In summary, we conclude that our algorithm, PASTA, reduces the number of iterations by at least $O\left(\frac{\text{vol}(R)^{\frac{3}{4}} \cdot \Delta(G)^{\frac{5}{4}}}{\epsilon}\right)$ compared to FDP, when a $(1 + \epsilon)$ -approximate solution is required. $\square$

B MORE DISCUSSIONS

B.1 Review of FISTA

We first briefly review the well-known PGD-based first-order optimization algorithm, fast iterative shrinkage-thresholding algorithm (FISTA) [9].

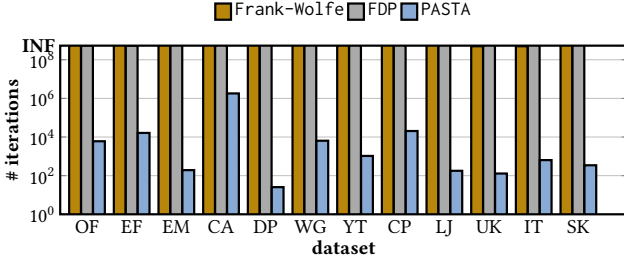
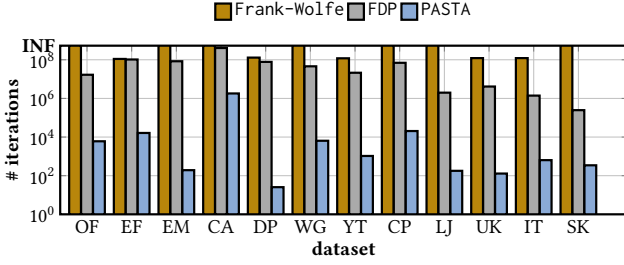
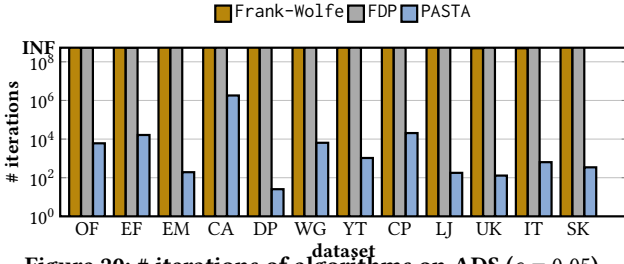
Algorithm 5: FISTA [9]

input : functions f and h , learning rate η , and iterations T
output : $x^{(T)}$
1 initialize $x^{(0)}$, s.t. $h(x^{(0)}) \leftarrow 0$;
2 $y^{(0)} \leftarrow x^{(0)}$;
3 **foreach** $t = 1, 2, \dots, T$ **do**
4 $x^{(t)} \leftarrow \text{prox}_h(y^{(t)} - \eta \cdot \nabla f(y^{(t)}))$;
5 $y^{(t)} \leftarrow x^{(t)} + \frac{t-1}{t+2} (x^{(t)} - x^{(t-1)})$;
6 **return** $x^{(T)}$;

As shown in Algorithm 5, the FISTA proceeds in an iterative manner (lines 3–5). In each iteration, it computes the gradient of the smooth function f , followed by a proximal step w.r.t. the non-smooth function h via gradient projection (line 4). FISTA is particularly efficient when the projection step can be computed at low cost.

B.2 The naive Fran-Wolfe-based ADS algorithm

The basic Frank-Wolfe algorithm can be adapted for the ADS problem, yielding the Frank-Wolfe-ADS version shown in Algorithm 1.

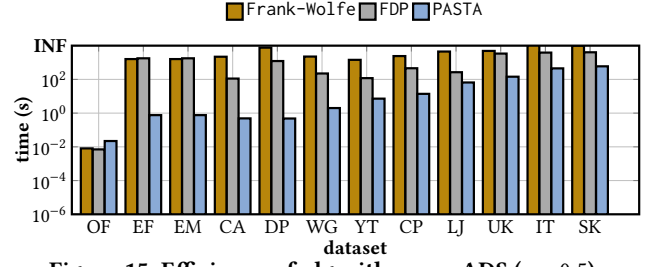
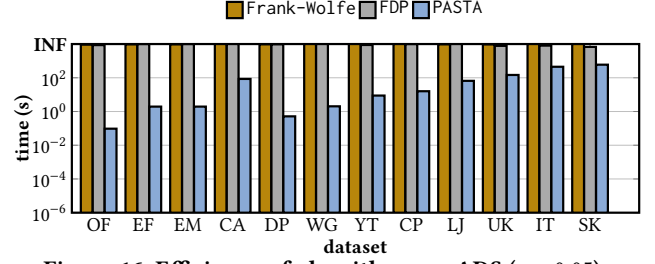
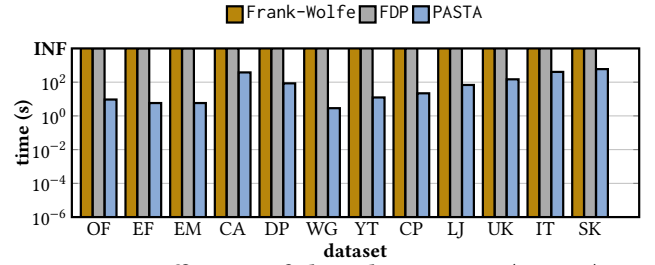
Figure 18: # iterations of algorithms on ADS ($\epsilon = 0.01$).Figure 19: # iterations of algorithms on ADS ($\epsilon = 0.5$).Figure 20: # iterations of algorithms on ADS ($\epsilon = 0.05$).**Algorithm 6:** Naive-Frank-Wolfe

input : Graph $G=(V, E)$, two vertex sets R, A and an integer T
output: An approximate ADS $\mathcal{D}$

- 1 **foreach** $(u, v) \in E$ **do** $\alpha_{u,v}^{(0)} \leftarrow \frac{1}{2}; \alpha_{v,u}^{(0)} \leftarrow \frac{1}{2};$
- 2 **foreach** $u \in V$ **do** $r^{(0)}(u) = \sum_{v \in N(u, G)} \alpha_{u,v}^{(0)};$
- 3 **foreach** $t \leftarrow 1, 2, 3, \dots, T$ **do**
- 4 $\eta_t \leftarrow \frac{2}{t+2};$
- 5 **for** $(u, v) \in E$ **do**
- 6 $\hat{\alpha}_{u,v} \leftarrow \alpha_{u,v}^{(t-1)} - \eta_t \cdot \nabla f(\alpha)_{u,v};$
- 7 $\hat{\alpha}_{v,u} \leftarrow \alpha_{v,u}^{(t-1)} - \eta_t \cdot \nabla f(\alpha)_{v,u};$
- 8 $\alpha^{(t)} \leftarrow (1 - \gamma_t) \cdot \alpha^{(t-1)} + \gamma_t \cdot \hat{\alpha};$
- 9 **foreach** $u \in V$ **do** $r^{(t)}(u) = \sum_{v \in N(u, G)} \alpha_{u,v}^{(t)};$

// Extract the ADS

- 10 sort all $v \in V$ by weight $r^{(T)}(v);$
- 11 $s^* \leftarrow \arg \max_{1 \leq i \leq n} \rho_R(V_i)$, where V_i is the top- i vertices in $V;$
- 12 **return** $G[V_{s^*}];$

Figure 15: Efficiency of algorithms on ADS ($\epsilon = 0.5$).Figure 16: Efficiency of algorithms on ADS ($\epsilon = 0.05$).Figure 17: Efficiency of algorithms on ADS ($\epsilon = 0.01$).**B.3 Relevant to the graph data management**

Indeed, the key contribution of our work is to propose an ADS discovery algorithm that is both theoretically and practically efficient. We would like to highlight that the ADS problem remains highly relevant to graph data management and graph database systems. First, the ADS solution has been used in many fundamental graph data management tasks [43], such as supporting graph visualization, community detection (or search), and identifying near-cliques. Second, modern graph database systems such as Neo4j and Nebula support community detection and community search operators, and ADS discovery is closely related to these functionalities. From another perspective, ADS discovery can be viewed as a graph query optimization task, where the goal is to identify relevant subgraphs from a large graph G given a query Q . Since current graph databases lack native support for ADS discovery, incorporating ADS discovery into modern systems presents an interesting and valuable future direction. Third, the DSD (including ADS) problem is a core task in graph data management and has attracted increasing attention in top-tier data management venues, with more than 20 papers published across SIGMOD, VLDB, PODS, ICDE, TODS, and VLDBJ [43].

B.4 Explanation of r and β vectors

We first clarify the overall meaning of the quantity $r(v) + \beta(v)$. For a vertex v , the value $r(v) + \beta(v)$ represents its overall contribution to the ADS. Intuitively, a larger value indicates that v is more likely to be included in the optimal ADS. Moreover, once the optimal solution is reached, all vertices in the ADS share the same value of $r(v) + \beta(v)$, which is guaranteed by Fujishige's theorem [29].

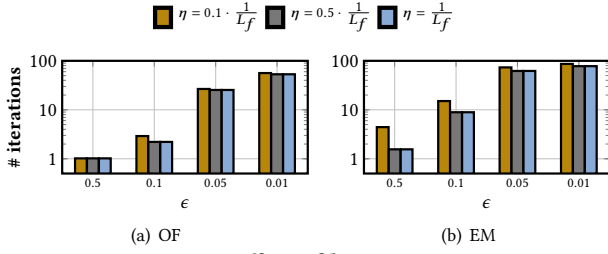
Figure 21: Effect of learning rate η .

Table 7: Statistics of datasets for bundle recommendation.

Dataset	# of vertices			# of edges	
	#User (U)	#Item (I)	#Bundle (B)	#U-I	#U-B
Netease	18,528	123,628	22,864	1,128,065	302,303
Youshu	8,039	32,770	4,771	138,515	51,377

Table 8: Performance of BGCN-R-ADS and BGCN-NR-ADS.

Dataset	Method	Recall@10	Recall@20	Recall@30	Recall@40
Netease	BGCN-R-ADS	0.0390	0.0647	0.0852	0.1044
	BGCN-NR-ADS	0.0404	0.0655	0.0865	0.1050
Youshu	BGCN-R-ADS	0.1765	0.2522	0.3088	0.3489
	BGCN-NR-ADS	0.1811	0.2638	0.3216	0.3633

Dataset	Method	NDCG@10	NDCG@20	NDCG@30	NDCG@40
Netease	BGCN-R-ADS	0.0214	0.0283	0.0329	0.0369
	BGCN-NR-ADS	0.0221	0.0288	0.0336	0.0374
Youshu	BGCN-R-ADS	0.1039	0.1274	0.1405	0.1489
	BGCN-NR-ADS	0.1049	0.1281	0.1422	0.1515

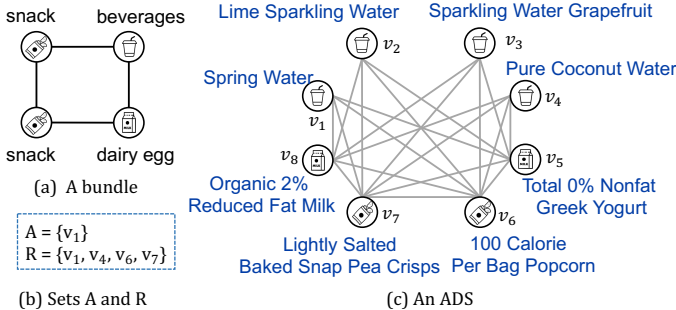


Figure 22: A case study on Instacart e-commerce network.

Specifically, for the optimal ADS $G[S]$ and every vertex $v \in S$, we have $r(v) + \beta(v) = \rho_R^*$. In addition, all vertices outside the ADS have strictly smaller values. This property directly motivates our extraction strategy in Section 3.1, where we select vertices with the largest values of $r + \beta$ as the candidate ADS.

We now explain the individual roles of r and β . The variable $r(v)$ captures the *intrinsic contribution* of vertex v to the NR-density objective. From an optimization perspective, a larger $r(v)$ indicates that including v helps increase the NR-density of the solution, independent of anchoring constraints. In contrast, $\beta(v)$ is an *adjustment variable* introduced to enforce the anchored set constraint. Its role is to bias the optimization toward satisfying the anchoring requirement by either encouraging or penalizing vertices in the anchored set A . Importantly, for any vertex $u \notin A$, we have $\beta^*(u) = 0$, meaning that non-anchored vertices are selected purely based on their intrinsic contribution $r(u)$.

Let (r^*, β^*) denote the optimal solution and S denote the ADS. Then the structure of $r^* + \beta^*$ can be summarized as follows:

- Vertices in the anchored set A attain the maximum value ρ_R^* due to the adjustment introduced by β^* .
- Vertices in $S \setminus A$ also attain the same maximum value, solely through r^* since $\beta^*(u) = 0$ for them.
- Vertices outside S have strictly smaller values of $r^* + \beta^*$.

Together, these properties explain both the meaning of r and β and why the combined quantity $r + \beta$ naturally characterizes membership in the ADS.

C ADDITIONAL EXPERIMENTS

In this section, we present the additional experimental results:

• **1. Efficiency of approximation algorithms.** We evaluate the efficiency of three approximations on all datasets with ϵ from 0.5 to 0.01. As shown in Figures 15 ~ 17, we can see that: (1) Our algorithm PASTA is up to four orders of magnitude faster than the baseline methods, since it has a better theoretical guarantee and can find the ADS over the very small graph. For example, on the OF dataset, PASTA only takes 10s to find the 1.01-approximation solution, while both FDP and Frank-Wolfe take more than three days. (2) On more than half of the datasets, neither Frank-Wolfe nor FDP can find solutions that meet the accuracy requirement within three days, which is consistent with our theoretical analysis.

• **2. The number of iterations.** In this experiment, we report the number of iterations that each algorithm needed to find the solutions with various accuracy requirements, i.e., ϵ from 0.5 to 0.01 in Figures 19 ~ 18. We make the following observations: (1) PASTA requires significantly fewer iterations to obtain approximate solutions compared to its competitors. For example, on the DP dataset, FW, FDP, and PASTA require around 129M, 77M, and 41 iterations, respectively. (2) As the accuracy requirement increases (i.e., ϵ decreases from large to small), the number of iterations required by FW and FDP grows significantly, whereas our method PASTA remains stable.

• **3. Effect of learning rate.** Indeed, our algorithm is not highly sensitive to the learning rate η . As shown in Figure 21, we report the number of iterations needed to achieve different ϵ values under various learning rates. The results indicate that the performance of PASTA remains stable.

• **4. Bundle recommendation.** The ADS can be used to enhance the performance of the state-of-the-art bundle product recommendation method BGCN [13], which aims to recommend a bundle of products to a single user. Specifically, BGCN [13] trains a Graph Neural Network (GNN)-based recommendation model using three networks, including the user-bundle interaction network, user-item interaction network, and bundle-item affiliation network, and then predicts whether a user will interact with a certain bundle. However, due to exposure bias and item diversity imbalance [13], the bundle-item affiliation network is often sparse and fails to capture accurate bundle-item relations. To alleviate the sparsity issue above, we propose to detect ADSs from the item co-purchasing network and then use these ADSs to augment the bundle-item affiliation network, thereby enhancing the performance of BGCN. Specifically, we apply PISTA ($\epsilon=0.1$) to detect ADSs by treating the target items as the anchored set (A), and the items that frequently co-occur with them in the same bundles as the reference set (R). Afterwards, in the bundle-item affiliation network, we add links between each bundle

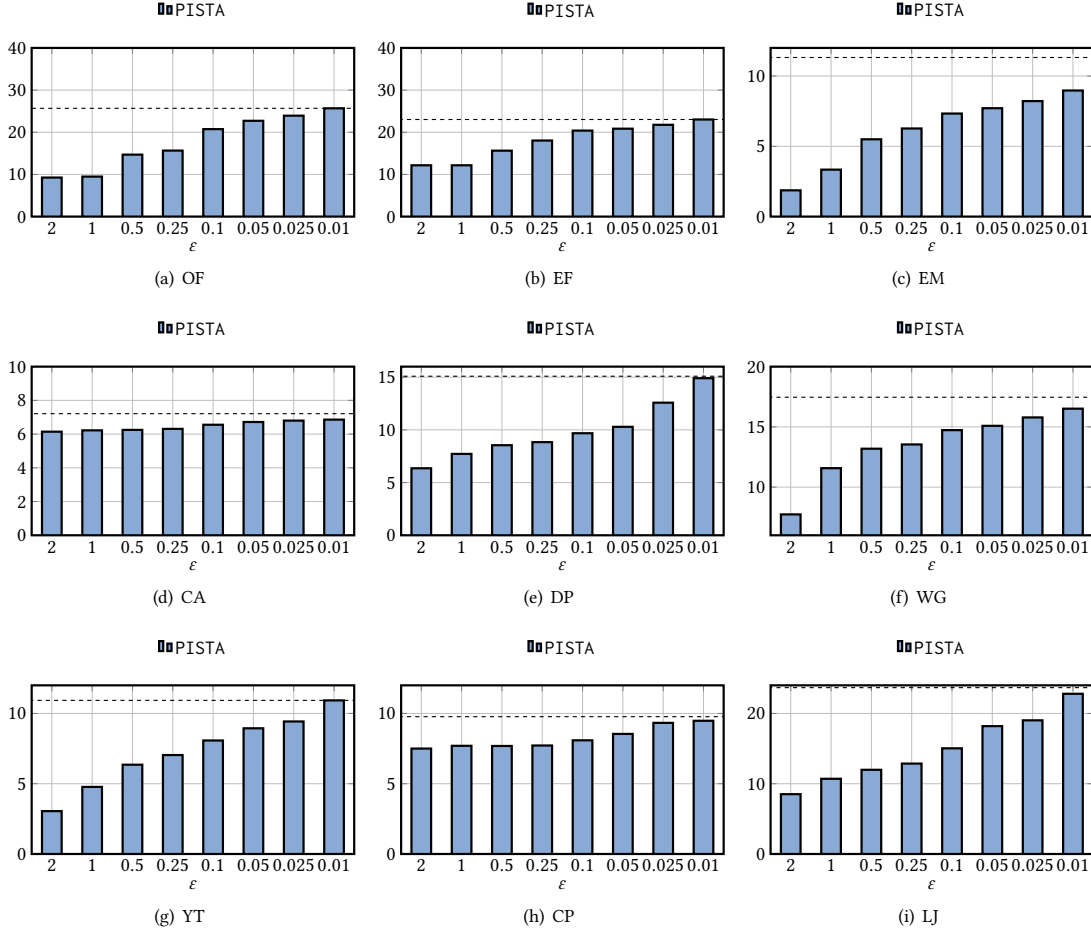


Figure 23: The NR-densities of the solutions obtained by PISTA under different ϵ values.

and the items that belong to the same ADSs, enabling BGCN to leverage richer high-quality item associations. Finally, we run BGCN on the augmented networks, and denote this augmented approach by BGCN-ADS. We experimentally compare BGCN and BGCN-ADS on two real-world datasets, namely Netease and Youshu, whose statistics are reported in Table 7. We adopt two widely used metrics, i.e., where Recall@K measures the ratio of test bundles that have been contained by the top-K ranking list, while NDCG@K complements Recall by assigning higher scores to the hits at higher positions of the list. As shown in Table 5, BGCN-ADS outperforms BGCN in terms of Recall@K and NDCG@K. This is because, it uses ADSs to augment the bundle-item affiliation network, which alleviates the sparsity issue by more precisely capturing the relationships between bundles and items. In addition, we show that ADS under R-density can also be used to enhance the BGCN model following the same idea. However, its performance is consistently inferior to that of ADS under NR-density, which motivates our focus on studying ADS discovery under NR-density.

• **5. Compared to other local densities.** As shown in prior work [21, 73], ADS obtained under NR-density (NR-ADS) consistently yield better effectiveness than those based on R-density (R-ADS), the most commonly used non-strongly local density. We

further validate this in two downstream applications — bundle recommendation and community search — using the same settings as in our Section 7. As shown in Table 8 and Figure 8, NR-ADS consistently outperforms R-ADS when used to augment the BGCN model, i.e., BGCN-NR-ADS achieves better performance than BGCN-R-ADS. Similarly, in community search, NR-ADS yields communities of higher quality than R-ADS. This advantage stems from the strong locality property of NR-density, which ensures that the discovered ADSs are more tightly centered around the anchored set. In contrast, R-density may include loosely related vertices, therefore degrading effectiveness.

• **6. Case study on bundle recommendation.** We use the Instacart e-commerce network, which contains a wide variety of products. We consider a real sales bundle with three types of items (i.e., “beverages”, “daily egg”, and “snack”), as shown in Figure 22(a). The intuition behind this bundle is to identify beverage and dairy/egg products that are frequently purchased together with snacks, enabling more effective product promotion. Figure 22(c) shows an ADS discovered by PISTA, where the anchored set A and reference set R are defined according to the sales bundle (see Figure 22(b)). The resulting ADS is highly cohesive and captures products that are strongly related and of interest to customers. Thus, ADS can be effectively used for product promotion. For instance, a bundle

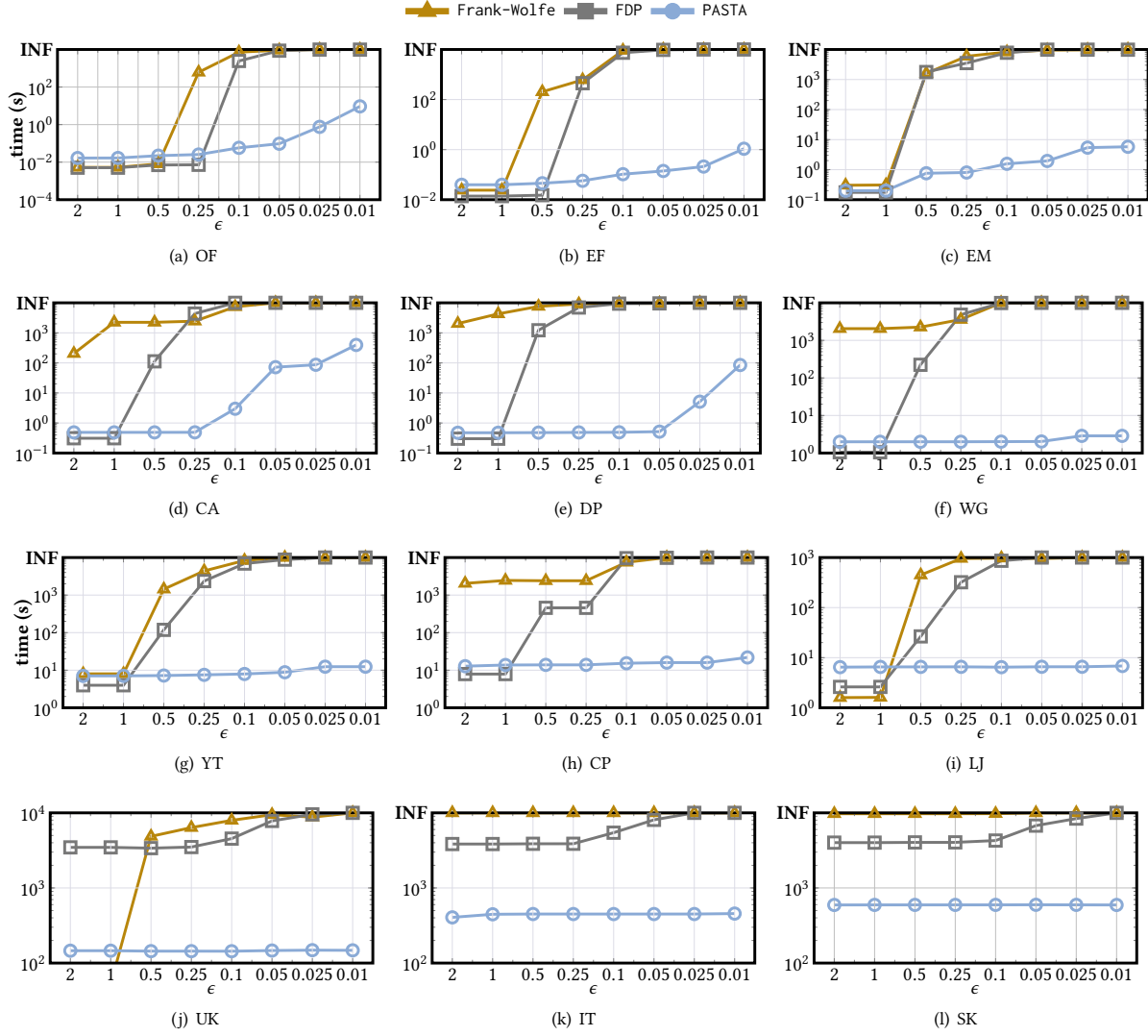


Figure 24: Efficiency results of $(1 + \epsilon)$ -approximation algorithms.

consisting of two snacks, one beverage, and one dairy/egg product extracted from the discovered ADS can be offered at a discounted price to improve sales.

• **7. Effect of ϵ .** In this experiment, we further evaluate PASTA under a wider range of ϵ values (from 2 to 0.01) and report the corresponding NR-densities achieved by PASTA. As shown in Figures 23 and 24, we make the following observations: 1) When low-accuracy solutions are sufficient (e.g., $\epsilon = 2$ and $\epsilon = 1$), both methods achieve

comparable performance on small datasets, while our method significantly outperforms on large datasets due to its effective graph reduction technique. 2) As ϵ decreases, the performance advantage of our method becomes increasingly pronounced, highlighting its stronger theoretical convergence guarantees. 3) On most datasets, setting $\epsilon = 0.1$ yields NR-densities that are very close to the optimal NR-density of ADS.